ELEC2104/9704 Tutorial 8 Solutions (2025)

1 Question 1

Find the Q-point for the circuit in the figure below for

(a) $\beta_F = 40$

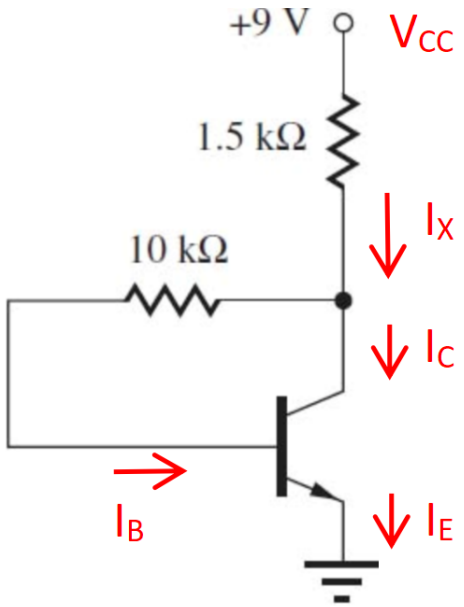
(b) $\beta_F = 120$

(c) $\beta_F = 250$

(d) $\beta_F = \infty$

(e) Find the Q-point in part (a) using the numerical iteration method if $I_S = 0.5$ fA and $V_T = 25.8$ mV.

(f) Find the Q-point in part (c) using the numerical iteration method if $I_S = 0.5$ fA and $V_T = 25.8$ mV.



Solution:

First, we write the emitter current as the sum of base and collector currents. Also, from KCL, we notice I_X is also the sum of base and collector currents.

$$I_X = I_E = I_B + I_C \tag{1}$$

Applying KVL around the loop including voltage source V_{CC}, collector resistor $R_C = 1.5$ kΩ, base resistor $R_B = 10$ kΩ, and base-emitter V_{BE}, we can write the equation:

$$V_{CC} - 1500I_X - 10000I_B - V_{BE} = 0 \tag{2}$$

From the property of BJT, we can relate collector current to base current using current gain:

$$I_C = \beta_F I_B \tag{3}$$

Then, we substitute $I_X = I_B + I_C = (1 + \beta_F)I_B$ into the KVL equation, we can solve for the base current.

$$V_{CC} - 1500(1 + \beta_F)I_B - 10000I_B - V_{BE} = 0 \tag{4}$$

$$I_B = \frac{V_{CC} - V_{BE}}{1500(1 + \beta_F) + 10000} \tag{5}$$

Finally, we write the expression for the collector current is a function of the using the current gain. We can find the collector current is a function of β_F, which is useful for part A, B, C, D.

$$I_C = \beta_F I_B = \frac{\beta(V_{CC} - V_{BE})}{1500(1 + \beta_F) + 10000} \tag{6}$$

1.1 Part A

Substitute $\beta_F = 40$ into the collector current expression:

$$I_C = \frac{\beta_F(V_{CC} - V_{BE})}{1500(1 + \beta_F) + 10000} = \frac{40(9 - 0.7)}{1500(1 + 40) + 10000} = 4.64\,\text{mA} \tag{7}$$

Next, compute the collector–emitter voltage. Since $V_E = 0$, we have:

$$V_{CE} = V_C - V_E = V_C = \left[9 - 1500I_X\left(1 + \tfrac{1}{\beta_F}\right)\right] = 1.87\,\text{V} \tag{8}$$

Where we used KVL to get $V_C = V_{CC} - 1.5\ \text{k}\Omega \cdot I_X$ and (1) and (3) give $I_X = I_E = I_C\left(1 + \frac{1}{\beta_F}\right)$.

Therefore, the Q-point of the transistor is:

$$Q = (I_C, V_{CE}) = (4.64\,\text{mA},\ 1.87\,\text{V}) \tag{9}$$

1.2 Part B

Take $\beta_F = 120$. The collector current is

$$I_C = \frac{\beta_F(V_{CC} - V_{BE})}{1500(1 + \beta_F) + 10000} = \frac{120(9 - 0.7)}{1500(1 + 120) + 10000} = 5.2\,\text{mA} \tag{10}$$

The collector–emitter voltage is

$$V_{CE} = \left[9 - 1500I_C\left(1 + \tfrac{1}{\beta_F}\right)\right] - 0 = 1.14\,\text{V} \tag{11}$$

Thus, the Q-point is

$$Q = (I_C, V_{CE}) = (5.2\,\text{mA},\ 1.14\,\text{V}) \tag{12}$$

1.3 Part C

Take $\beta_F = 250$. The collector current is

$$I_C = \frac{250(9 - 0.7)}{1500(1 + 250) + 10000} = 5.37\,\text{mA} \tag{13}$$

The collector–emitter voltage is

$$V_{CE} = \left[9 - 1500I_C\left(1 + \tfrac{1}{\beta_F}\right)\right] - 0 = 0.91\,\text{V} \tag{14}$$

Thus, the Q-point is

$$Q = (I_C, V_{CE}) = (5.37\,\text{mA},\ 0.91\,\text{V}) \tag{15}$$

1.4 Part D

Take $\beta_F = \infty$. Then

$$I_C = \frac{\infty(9-0.7)}{1500(1+\infty)+10000} = \frac{9-0.7}{1500} = 5.53\,\text{mA} \tag{16}$$

The collector–emitter voltage is

$$V_{CE} = \left[9 - 1500 I_C \left(1 + \tfrac{1}{\infty}\right)\right] - 0 = 0.705\,\text{V} \tag{17}$$

Thus, the Q-point is

$$Q = (I_C, V_{CE}) = (5.53\,\text{mA},\ 0.705\,\text{V}) \tag{18}$$

1.5 Part E

The collector current is expressed by the exponential model:

$$I_C = I_S \exp\left(\frac{V_{BE}}{V_T}\right) \tag{19}$$

Thus, the first expression for V_{BE} is

$$V_{BE1} = V_T \ln\left(\frac{I_C}{I_S}\right) \tag{20}$$

Apply KVL to the circuit, we get the second expression for V_{BE} is

$$V_{BE2} = 9 - 1500 \times (I_C + I_B) - 10000 \times (I_B) = 9 - 1500\left(1 + \frac{1}{\beta_F}\right) I_C - 10000\left(\frac{I_C}{\beta_F}\right) \tag{21}$$

Now we construct the function

$$f(I_C) = V_{BE1} - V_{BE2} = \left[V_T \ln\left(\tfrac{I_C}{I_S}\right)\right] - \left[9 - 1500\left(1 + \tfrac{1}{\beta_F}\right) I_C - 10000\left(\tfrac{I_C}{\beta_F}\right)\right] \tag{22}$$

The derivative of the function is

$$f'(I_C) = \frac{V_T}{I_C} + 1500\left(1 + \tfrac{1}{\beta_F}\right) + 10000 \cdot \left(\tfrac{1}{\beta_F}\right) \tag{23}$$

Newton's iteration formula is

$$I_C^{(n+1)} = I_C^{(n)} - \frac{f(I_C^{(n)})}{f'(I_C^{(n)})} \tag{24}$$

For $\beta_F = 40$, after several iterations, we can solve

$$I_C \approx 4.60\,\text{mA} \tag{25}$$

Thus, the Q-point is

$$Q = (I_C, V_{CE}) = (4.6\,\text{mA},\ 1.92\,\text{V}) \tag{26}$$

1.6 Part F

Using the same working procedure as part E, we take $\beta_F = 250$, we iterate the equation (45) for several iterations than we can get

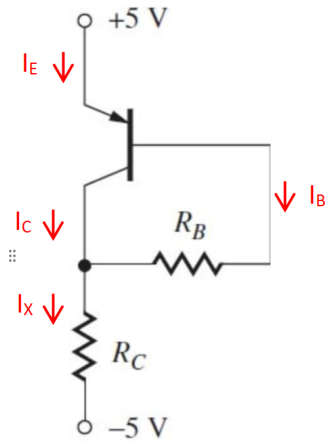
$$I_C \approx 5.32\,\text{mA} \tag{27}$$

Thus, the Q-point is

$$Q = (I_C, V_{CE}) = (5.32\,\text{mA},\ 0.987\,\text{V}) \tag{28}$$

2 Question 2

Design the bias circuit in the figure below to give a Q-point of $I_C = 10\,\text{mA}$ and $V_{EC} = 3\,\text{V}$ if the transistor current gain $\beta_F = 60$. What is the Q-point if the current gain of the transistor is actually 40?



Solution:

2.1 Part A

Note that PNP type BJT! The current directions and voltage polarities are flipped (vs. NPN BJT).

The base current is obtained from the given collector current using $I_C = 10\,\text{mA}$ and $\beta_F = 60$:

$$I_B = \frac{I_C}{\beta_F} = \frac{10\,\text{mA}}{60} = 0.167\,\text{mA} \tag{1}$$

The emitter-collector (PNP) voltage is specified as $V_{EC} = 3\,\text{V}$. Ohm's Law across the resistor R_C gives (note the negative voltage source $V_{CC} = -5\,\text{V}$):

$$\begin{aligned} \frac{V_C - (-5\,\text{V})}{R_C} &= I_X \\ R_C &= \frac{V_C + 5\,\text{V}}{I_X} \end{aligned} \tag{2}$$

By KVL, $V_C = 5\,\text{V} - V_{EC}$, which includes $V_{EE} = +5\,\text{V}$, the voltage source at the top of the circuit. Since the question states $V_{EC} = 3\,\text{V}$, we then have $V_C = 5\,\text{V} - 3\,\text{V} = 2\,\text{V}$. Thus, our expression for R_C becomes:

$$R_C = \frac{2\,\text{V} + 5\,\text{V}}{I_X} = \frac{7\,\text{V}}{I_X} \tag{3}$$

Now, by KCL, $I_X = I_C + I_B$, which are all known quantities. Thus, we can solve for R_C:

$$R_C = \frac{7\,\text{V}}{(0.01 + 0.000167)\,\text{A}} = 688.5\,\Omega \tag{4}$$

Since this is design task, we need to pick a existing resistor value in real life, so we choose $R_C = 680\,\Omega$. To choose the base resistor, make the assumption that the BJT is on, so $V_{EB} \approx 0.7\,\text{V}$. Using Ohm's

Law and the fact from KVL that $V_{BC} = V_{EC} - V_{EB}$:

$$\begin{aligned} R_B &= \frac{V_{BC}}{I_B} \\ &= \frac{V_{EC} - V_{EB}}{I_B} \\ &= \frac{3\,\text{V} - 0.7\,\text{V}}{0.167\,\text{mA}} \\ &= 13.77\,\text{k}\Omega \end{aligned} \tag{5}$$

Again, since this is design task, we need to pick a existing resistor value in real life, so we choose $R_B = 14\,\Omega$.

2.2 Part B

Now we apply the design choices $R_B = 14\,\text{k}\Omega$, $R_C = 680\,\Omega$, and take $\beta_F = 40$.

Applying KVL from the top $+5\,\text{V}$ voltage source, through the base V_{EB} and base resistor R_B, down through the R_C resistor to the bottom negative voltage source $-5\,\text{V}$:

$$5\,\text{V} - V_{EB} - R_B I_B - R_C I_X - (-5\,\text{V}) = 0 \tag{6}$$

Note the V_{EB} voltage polarity for the PNP. Also note that (KCL) $I_X = I_B + I_C = (1 + \beta_F) I_B$.

Substituting $V_{BE} \approx 0.7\,\text{V}$, we solve for base current:

$$\begin{aligned} 5\,\text{V} - 0.7\,\text{V} - 14\,\text{k}\Omega I_B - 680\Omega(1 + 40) I_B + 5\,\text{V} &= 0 \\ (41880\,\Omega) I_B &= 9.3\,\text{V} \\ I_B &= 222.1\,\mu\text{A} \end{aligned} \tag{7}$$

Then, we determine the collector current.

$$I_C = \beta_F I_B = 40 \cdot 222.1\,\mu\text{A} = 8.88\,\text{mA} \tag{8}$$

Next, we can determine the collector–emitter voltage using

$$V_{EC} = V_E - V_C = 5\,\text{V} - \Big[R_C(1 + \beta_F) I_B + (-5)\Big] = 3.81\,\text{V} \tag{9}$$

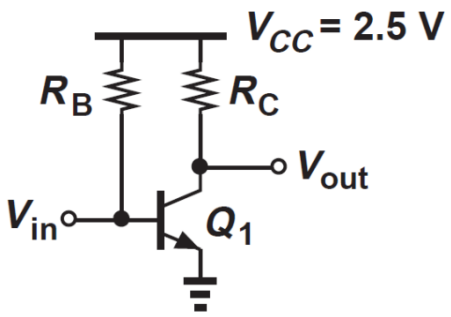
Where in the above we used KVL ($V_C - R_C I_X = -5\,\text{V}$) and KCL ($I_X = I_B + I_C = (1 + \beta_F) I_B$).

Therefore, the Q-point is

$$Q = (I_C, V_{EC}) = (8.88\,\text{mA},\, 3.81\,\text{V}) \tag{10}$$

3 Question 3

Design the Common-Emitter amplification stage of Fig. 5.161 for a power budget of 1 mW and a voltage gain of 20. Assume $\beta = 100$, $V_T = 26\,\text{mV}$, and $I_s = 6 \times 10^{-16}\,\text{A}$.



Solution:

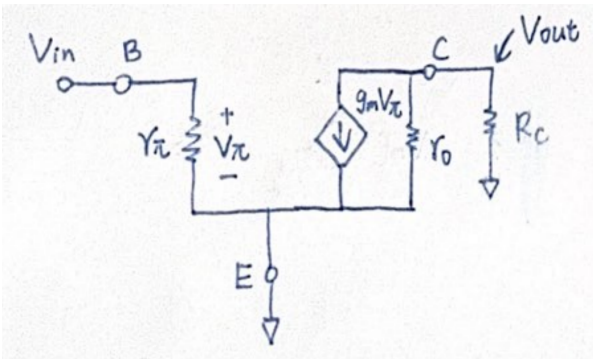
The total power dissipated is written below. Since I_B is very small compared to I_C, we ignore I_B.

$$P = V_{CC} \cdot I_{\text{total}} = V_{CC} \cdot (I_B + I_C) \approx V_{CC} \cdot I_C \tag{1}$$

Thus, rearranging (1), we can solve the collector current.

$$I_C = \frac{P}{V_{CC}} = \frac{1\,\text{mW}}{2.5\,\text{V}} = 0.4\,\text{mA} \tag{2}$$

For the voltage gain, we start from drawing the small signal circuit diagram. Since the early effect is not mentioned, r_o is ∞.



According to the circuit diagram, we can write the expression for the voltage gain A_v.

$$A_V = \frac{v_{out}}{v_{in}} = \frac{-g_m v_\pi R_C}{v_\pi} = -g_m R_C = -\frac{I_C}{V_T} R_C \tag{3}$$

Since A_v, I_C and V_T, we can rearrange (3) and solve for R_C:

$$R_C = \left| \frac{A_V V_T}{I_C} \right| = \left| \frac{-20 \cdot 0.026}{0.4 \times 10^{-3}} \right| = 1300\,\Omega \tag{4}$$

For the design problem, $R_C = 1300\,\Omega$ is possible.
Now applying KVL at the input loop, we can write

$$V_{CC} - I_B R_B - V_{BE} = 0 \tag{5}$$

Since $I_B = I_C/\beta_F$ and $V_{BE} = V_T \ln(I_C/I_S)$, this gives:

$$V_{CC} - \frac{I_C}{\beta_F} R_B - V_T \ln\left(\frac{I_C}{I_S}\right) = 0 \tag{6}$$

Substitute $V_{CC} = 2.5\,\text{V}$, $V_T = 0.026\,\text{V}$, $I_C = 0.4\,\text{mA}$, and $I_S = 6 \times 10^{-16}\,\text{A}$:

$$2.5 - \frac{I_C}{\beta_F} R_B - (0.026) \ln\left(\frac{I_C}{6 \times 10^{-16}}\right) = 0 \tag{7}$$

Solving for R_B gives

$$R_B = 448\,\text{k}\Omega \tag{8}$$

For the design problem, $R_B = 448\,\text{k}\Omega$ is possible.

4 Question 4

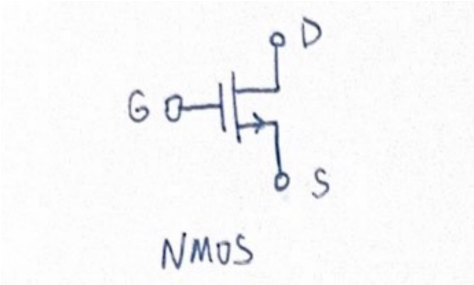
Find the region of operation and drain current in an NMOS transistor with $\mu_n C_{\text{ox}} = 200\,\mu\text{A/V}^2$, $W/L = 10/1$, and $V_{TN} = 0.75\,\text{V}$ for:

(a) $V_{GS} = 2\,\text{V}$ and $V_{DS} = 2.5\,\text{V}$,

(b) $V_{GS} = 2\,\text{V}$ and $V_{DS} = 0.2\,\text{V}$,

(c) $V_{GS} = 0\,\text{V}$ and $V_{DS} = 4\,\text{V}$.

(d) Repeat parts (a)–(c) for $\mu_n C_{\text{ox}} = 300\,\mu\text{A/V}^2$.



Solution:

The typical working procedure to solve an MOSFET problem is

1. Check operating region.
2. Calculate I_D based on the region.

4.1 Part A

Step 1. check the operating region
According to the calculation below, we can find that the NMOS operates in the saturation region.

$$V_{GS} - V_{TN} = 2 - 0.75 = 1.25\,\text{V} < V_{DS} = 2.5\,\text{V}$$

Step 2. Calculate I_D based on the region.
Next we apply the formula for the saturation I_D.

$$I_D = \tfrac{1}{2}(\mu_n C_{ox})\left(\tfrac{W}{L}\right)(V_{GS} - V_{TN})^2 = \tfrac{1}{2}(200 \times 10^{-6})(\tfrac{10}{1})(2 - 0.75)^2 = 1.56\,\text{mA} \tag{1}$$

4.2 Part B

Step 1. Check the operating region
We compare $V_{GS} - V_{TN}$ with V_{DS}:

$$V_{GS} - V_{TN} = 1.25\,\text{V} > V_{DS} = 0.2\,\text{V}$$

Therefore, the NMOS operates in the **triode (linear) region**.
Step 2. Calculate I_D based on the region.
For the linear region, the drain current is

$$I_D = (\mu_n C_{ox})\left(\tfrac{W}{L}\right)\left[(V_{GS} - V_{TN} - \tfrac{V_{DS}}{2})V_{DS}\right] \tag{2}$$

Substituting values:

$$I_D = (200 \times 10^{-6})\left(\tfrac{10}{1}\right)\left[(2 - 0.75 - \tfrac{0.2}{2})(0.2)\right] = 0.46\,\text{mA}$$

4.3 Part C

Step 1. Check the operating region
We evaluate the gate-to-source condition:

$$V_{GS} - V_{TN} = 0 - 0.75 = -0.75 < 0$$

Thus, the NMOS is in the **cutoff region**.
Step 2. Calculate I_D based on the region.
Since the device is in cutoff,

$$I_D = 0 \tag{3}$$

4.4 Part D

Here we consider the same NMOS configuration, but with a different mobility–oxide capacitance product ($\mu_n C_{ox}$). The operating region remains the same as in parts (a)–(c), but the drain current I_D will scale with ($\mu_n C_{ox}$).

(a) Saturation Region

$$I_D \propto (\mu_n C_{ox})$$

$$I_D = \left(\frac{300}{200}\right)(1.56\,\text{mA}) = 2.34\,\text{mA} \tag{4}$$

(b) Triode / Linear Region

$$I_D \propto (\mu_n C_{ox})$$

$$I_D = \left(\frac{300}{200}\right)(0.46\,\text{mA}) = 0.96\,\text{mA} \tag{5}$$

(c) Cutoff Region

Since $V_{GS} < V_{TN}$,

$$I_D = 0 \tag{6}$$